MD ZAWADUL AMAN HREDOY

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ACADEMIC PROFILE

Computer Science and Engineering final-year undergraduate (specialising in Data Science) and Junior Software Engineer, combining production full-stack engineering expertise with academic research in Natural Language Processing and on-device LLM guardrails. Proven track record in building distributed queue architectures, concurrency-safe scheduling engines, and AI-driven SaaS platforms using modern TypeScript, React, and Node.js. Seeking admission to MSc programmes in Computer Science, Artificial Intelligence, or Data Science across the United Kingdom.

EDUCATION

United International University (UIU) — Dhaka, Bangladesh | 2022 – 2026

Bachelor of Science in Computer Science and Engineering

- **Specialisation:** Data Science
- **Academic Standing:** 78% (Marks)
- **Key Academic Modules:** Machine Learning, Artificial Intelligence, Data Mining, Database Management Systems, Computer Networks, Software Engineering, Computer Security, Systems Analysis & Design, Green Computing.

Savar Govt. College — Dhaka, Bangladesh | 2017 – 2019

Higher Secondary Certificate (HSC) — Science

- **Score:** 4.92 / 5.00

PROFESSIONAL EXPERIENCE

Ads Pillar — Dhaka, Bangladesh | August 2026 – Present

Junior Software Engineer

- Assisted in the full-stack architecture, development, testing, and deployment of client-facing and internal software solutions using modern JavaScript, TypeScript, and full-stack frameworks.
- Collaborated with senior software engineers to refactor data-fetching pipelines, optimise API throughput, and enhance application performance and user experience.
- Debugged and resolved technical issues across staging and production environments, enforcing structured error logging, input validation, and defensive programming practices.
- Participated actively in engineering sprint rituals, code reviews, and technical documentation to ensure maintainable architecture and strict adherence to software standards.

RESEARCH & ACADEMIC PROJECTS

An Autonomous Self-Healing LLM Guardrail System on Edge Hardware — Undergraduate Research Project | 2026

- Researched and designed an on-device safety guardrail framework for Large Language Models (LLMs) executing on resource-constrained edge hardware.
- Investigated self-healing intervention pipelines to intercept hallucinations, prompt injections, and unsafe token streams with minimal inference latency.
- Explored model quantization and edge runtimes to analyse trade-offs between memory footprint, computational overhead, and real-time generation fidelity.

Automatic Detection of Depression in Social Media Posts using NLP — Undergraduate Research Project | 2025 – 2026

- Formulated an end-to-end NLP classification pipeline to identify linguistic biomarkers and affective cues indicative of depressive sentiment in unstructured social media corpora.
- Engineered text preprocessing workflows (tokenisation, lemmatisation, stop-word mitigation) and extracted semantic representations for supervised and transformer-based classifiers.
- Benchmarked classification efficacy using precision, recall, and F1-score to assess predictive reliability and model generalisation.

TECHNICAL & SOFTWARE ENGINEERING PROJECTS

PlanPost — AI Content & Social Media Automation Platform — Live Platform | GitHub

Tech Stack: React, Node.js, Express, WebSockets, MongoDB, Redis, IndexedDB

- Engineered a distributed multi-platform publishing platform featuring automated post scheduling, queue management, and real-time event synchronization.
- Integrated generative AI copywriting pipelines to dynamically synthesize, format, and optimize campaign text and hashtags across diverse social networks.
- Developed **"Aristro"**, a visual post creation studio and canvas editor for composing and formatting rich social media creatives.
- Integrated advertising engine modules for campaign parameter configuration, automated budget scheduling, and multi-channel asset distribution.
- Built a real-time team messaging engine and collaborative review/approval workflow, enabling seamless communication between campaign managers and stakeholders.
- Implemented an offline-first caching layer utilizing browser **IndexedDB** to persist media assets locally and reduce redundant network payload transfers.

Smart CV — AI Career Advisor & Mentorship Platform Live Platform | GitHub

Tech Stack: Next.js 16 (App Router), React 19, TypeScript, Supabase, PostgreSQL, OpenAI API, Tailwind CSS, Zustand

- Architected an AI-powered career advisory SaaS platform featuring an interactive candidate dashboard, activity timeline, and version-controlled CV generation history.
- Implemented automated profile optimisation engines tailored for **LinkedIn** and **BDJobs**, utilizing the OpenAI API to reformat raw career records into ATS-compliant summaries and STAR-aligned achievements.
- Engineered a multi-parameter mentor discovery engine with dynamic filtering across expertise domains, pricing tiers, and real-time calendar availability.
- Developed a race-safe 4-step consultation booking system integrating Google Meet APIs, automated slot reservation, resume attachment streaming, guest invitations, and instant confirmation triggers.
- Integrated tiered subscription billing (30-day access passes), coupon validation, and checkout callback lifecycle handling.
- Implemented bilingual localisation (English and Bengali) with persistent cookie-based state and hydration mismatch mitigation.

HR Management & Consultation Booking Platform GitHub Client | GitHub Server

Tech Stack: Node.js 22, Express 5, TypeScript, React 19, MongoDB, Redis, BullMQ, Google API

- Engineered a high-concurrency booking platform with DST-safe availability slot generation across IANA timezones using **Luxon**.
- Eliminated concurrent double-booking race conditions without heavy transaction locks by designing sparse unique `slotKey` indexing coupled with post-insert overlap sweeps.
- Integrated Google Calendar and Google Meet APIs via OAuth 2.0 with **AES-256-GCM** encrypted token storage (`scrypt`-derived keys) and signed JWT state parameters for CSRF mitigation.
- Implemented asynchronous background job queues using **Redis and BullMQ** for automated 30-minute reminder emails, lifecycle notifications, and session state transitions.
- Maintained production-grade quality with **113+ automated integration and unit tests** using **Vitest** and **Supertest**; implemented Role-Based Access Control (RBAC) and structured Pino logging.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Aspire Leader Program — Aspire Institute (Founded by Harvard Faculty) Sep 2024

- Selected for an international leadership development fellowship for first-generation university scholars.
- Participated in interactive masterclasses led by faculty from Harvard, MIT, and Oxford, analysing emerging technological trends, economic systems, and global challenges.
- Collaborated within an international peer cohort to resolve complex case studies using systems-thinking and structured problem-solving frameworks.

UIU Computer Club — United International University General Member

- Supported logistical operations and planning for university-wide technical workshops, coding bootcamps, and competitive programming seminars.

Specialised Workshops & Certifications 2025

- **Generative AI Mastermind (Outskill):** Hands-on training in agentic workflows, custom GPT construction, and multi-step AI automation pipelines (Mar 2025).
- **Advanced Full-Stack Web Development:** Modern component architecture, state management, and asynchronous API design (Dec 2025).

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, Java, C, C++, PHP, SQL, HTML5, CSS3
- **Frontend Development:** React.js (v19), Next.js 16 (App Router), Tailwind CSS, TanStack Query, Zustand, React Hook Form, Vite
- **Backend & Distributed Systems:** Node.js, Express.js, RESTful APIs, WebSockets (Socket.IO), Redis, BullMQ, Microservices Architecture
- **Machine Learning & AI:** Natural Language Processing (NLP), Large Language Models (LLMs), Prompt Engineering, OpenAI API, Hugging Face, Scikit-learn
- **Databases & Cloud Storage:** MongoDB (Mongoose), PostgreSQL (Supabase), Redis (ioredis), IndexedDB
- **DevOps, Testing & Security:** Git, GitHub, Docker, Vitest, Supertest, Postman, Zod, JWT, OAuth 2.0, AES-256-GCM, Linux/Bash